

03 . 1

In fruit flies, the genes for body colour and wing length are linked. Explain what this means.

[1 mark]

A scientist investigated linkage between the genes for body colour and wing length. He carried out crosses between fruit flies with grey bodies and long wings and fruit flies with black bodies and short wings.

Figure 2 shows his crosses and the results.

- **G** represents the dominant allele for grey body and **g** represents the recessive allele for black body.
- **N** represents the dominant allele for long wings and **n** represents the recessive allele for short wings.

Figure 2

<i>Phenotype of parents</i>	grey body, long wings	×	black body, short wings
<i>Genotype of parents</i>	GGNN		ggnn
<i>Genotype of offspring</i>	GgNn		
<i>Phenotype of offspring</i>	all grey body, long wings		

These offspring were crossed with flies homozygous for black body and short wings.

The scientist's results are shown in **Figure 3**.

Figure 3

	GgNn	crossed with		ggnn
	Grey body, long wings	Black body, short wings	Grey body, short wings	Black body, long wings
Number of offspring	975	963	186	194

0 3 . 2 Use your knowledge of gene linkage to explain these results.

[4 marks]

[Extra space]

0 3 . 3 If these genes were **not** linked, what ratio of phenotypes would the scientist have expected to obtain in the offspring?

[1 mark]

0 3 . 4 Which statistical test could the scientist use to determine whether his observed results were significantly different from the expected results?

Give the reason for your choice of statistical test.

[2 marks]
